

JAMES RUSE AGRICULTURAL HIGH SCHOOL

2 Unit Term 2 Assessment 1998

Open Book Assessment

Time Allowed : 85 mins

All questions are of equal value

All necessary working to be shown.

SECTION A (Start a new page)

- (a) A particle moves in a straight line in such a way that its distance in metres from the origin after t seconds is given by

$$x = 2t^3 + 3t^2 - 36t + 10$$

- In which direction is the particle moving initially? Give reason.
 - When and where is the particle instantaneously at rest?
 - What total distance has the particle travelled in the first three seconds of its motion?
- (b) A box contains 5 white balls, 3 black balls and 2 red balls. Two balls are drawn at random. What is the probability that:-
- Both balls are white
 - One ball is white and the other is black
 - Both are the same colour.

SECTION B (Start a new page)

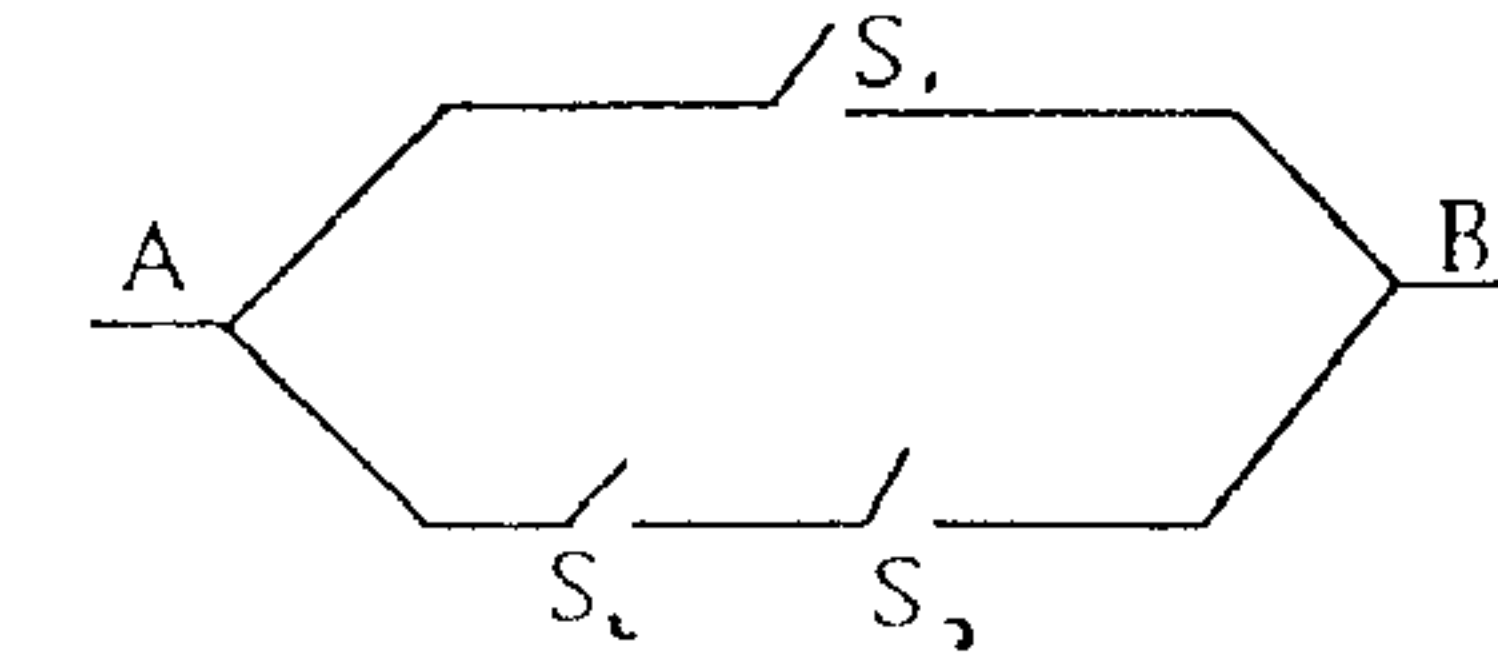
- (a) Find the quadratic equation whose roots are -3 and $\frac{2}{3}$ (answer to be written in the form $ax^2 + bx + c = 0$).
- (b) Given α and β are the roots of the equation $x^2 - 8x + 5 = 0$, find the value of:
- $\alpha + \beta$
 - $\alpha\beta$
 - $\frac{2}{\alpha} + \frac{2}{\beta}$
 - $\alpha^2 + \beta^2$
- (c) Solve $8 - 2x - x^2 > 0$
- (d) Determine whether $2x^2 - 3x + 2$ is positive definite, negative definite or indefinite. Give reasons.

SECTION C (Start a new page)

- (a) (i) The rate of growth of the population N of a city at any time t is directly proportional to the population at that time, thus $dN/dt = kN$. Show that after time t , then $N = N_0 e^{kt}$ satisfies the equation $dN/dt = kN$
- (ii) (a) If at the beginning of 1997, the population of a city is 1 000 000 and the population is increasing by 4% per annum, find the values of N_0 and k giving k to 4 dec places.
- (b) Hence determine the expected population of the city at the start of 2007 to the nearest thousand.
- (c) In what year would you expect the population to first exceed 2 000 000

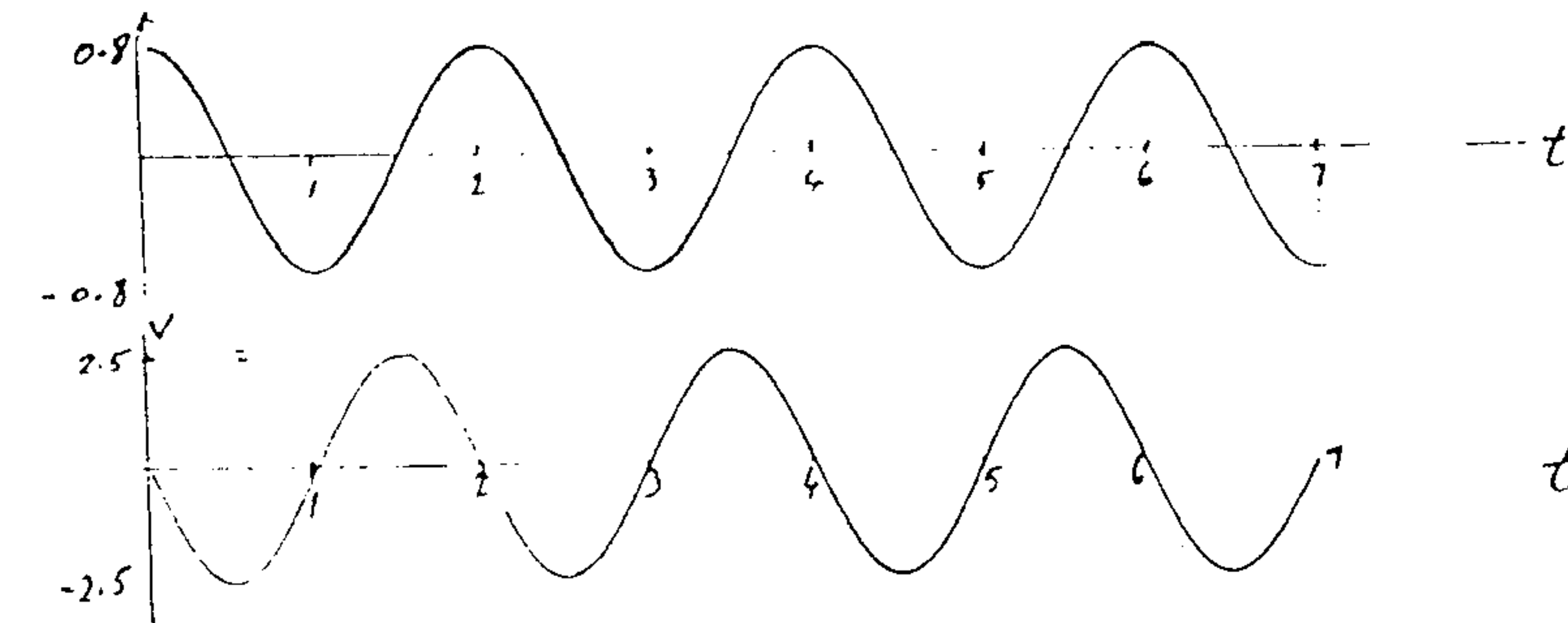
- (b) The diagram shows a switching circuit. The probability that any switch is open is $\frac{1}{4}$

Determine the probability that the current cannot flow from A to B



SECTION D (Start a new page)

- (a) A particle moves in a straight line. The diagrams below are the displacement and velocity graphs of the particle where x is in metres and t in seconds.



- Find the initial displacement and velocity.
 - State the time (s) when the particle has greatest speed.
 - Where does the particle first change direction?
 - State the time (s) when the particle has zero acceleration?
- (b) Two cards are selected at random from five cards bearing the numbers 2, 2, 4, 5, 6. Find the probability that the numbers on the cards:
- are different
 - have a sum of 9
 - have a difference of 1

SECTION E (Start a new page)

- (a) The acceleration of a particle moving in a straight line is given by $a = \frac{-1}{t^2}$, where the displacement (x) is measured in cm and the time (t) in seconds. If the particle is moving with velocity 6 cm/s, when $t = 1$ find the limiting velocity of the particle.
- (b) The line $y = x + k$ is drawn on the same diagram as the circle $x^2 + y^2 = 16$. Find the values of k that make this line
- touch the circle
 - intersect the circle in two points.
- (c) The line $x + 2y - 3 = 0$ forms the chord AB of the parabola $y = x^2 - 2$. Find the midpoint of AB.

END OF PAPER.

(98) $x = 2t^3 + 3t^2 - 36t + 110$

(i) $V = 6t^2 + 6t - 36$
 when $t=0$, $V = -36$
 $\therefore$ Moving left (Neg Vel) ①

(ii) $V = 6(t+3)(t-2)$
 $\therefore$ comes to rest after $t=2$ ①
 when $x = 2.8 + 3.4 - 72 + 110$
 $= -34 \text{ m.}$ ①

(iii) $t=3$ $x = 2.27 + 27 - 108 + 110$
 $= -17$ ①
 $\therefore$ DNT Travel
 $10 + 34 + 34 - 17 = 61 \text{ m.}$

(i) $P(WW) = \frac{5}{10} \cdot \frac{4}{9} = \frac{2}{9}$ ②

(ii) $P(WB) + P(BW) = 2 \times \frac{5}{10} \cdot \frac{3}{9} = \frac{1}{3}$ ②

(iii) $P(BB) + P(WW) + P(RR)$
 $= \frac{3}{10} \cdot \frac{2}{9} + \frac{5}{10} \cdot \frac{4}{9} + \frac{2}{10} \cdot \frac{1}{9}$
 $= \frac{28}{90}$
 $= \frac{14}{45}$ ②

L.B.

1) $(x+3)(x-\frac{2}{3}) = 0$ ①

$(x+3)(3x-2) = 0$
 $3x^2 + 7x - 6 = 0$ ①

$\therefore$ (i) $x+B = 8$ ①

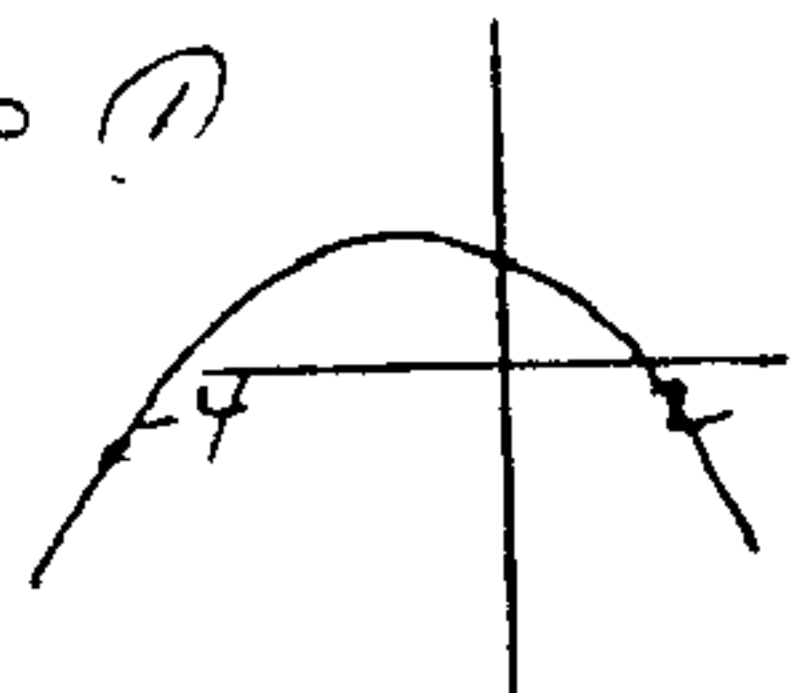
(iii) $\frac{2(x+B)}{2B} = \frac{10}{5}$ (2)

(iv) $x^2+B^2 = (x+B)^2 - 2xB$
 $= 14 - 2 \cdot 15$
 $= 54$ ②

(c) $-(x^2+2x-1) > 0$

$-(x+4)(x-2) > 0$ ①

$-4 < x < 2$ ①



(d) $2x^2 - 3x + 2$
 $\Delta = 9 - 4 \cdot 2 \cdot 2$ ①
 $= -7$
 < 0 & $a > 0$
 $\therefore$ position definite. ①

Section C.

(a) (i) $N = N_0 e^{kt}$
 $\frac{dN}{dt} = k N_0 e^{kt}$
 $= k N$... since $N = N_0 e^{kt}$
 (ii) $N_0 = 1000000$
 $1.04 N_0 = N_0 e^{kt}$
 $k = 0.0392 \approx 4\%$

(B) $t=10$
 $N = 1000000 e^{10 \cdot k}$
 $= 1000000 e^{0.392}$
 ≈ 1480000 (nearest thousand) ①

(Y) $2000000 = 1000000 e^{k \cdot 10.4}$
 $t \approx 17.7$ ①
 $1997 + 17 = 2014$ ①

(b)

$P(S_1 \text{ open}, S_2 \text{ open}, S_3 \text{ open})$

$+ P(S_1 \text{ open}, S_2 \text{ closed}, S_3 \text{ open})$ ②

$+ P(S_1 \text{ open}, S_2 \text{ open}, S_3 \text{ closed})$

$= \frac{1}{64} + 2 \left(\frac{1}{4} \times \frac{1}{4} \times \frac{3}{4} \right)$
 $= \frac{7}{64}$

Section D.

a (i) Initial displacement = 0.8 m ①
 Initial vel = 0 m/s ①

(ii) $\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}, \frac{11}{2}, \frac{13}{2}$ periods. ③

(iii) $x = -0.8 \text{ m.}$ ②

(iv) same as (ii). ②

41.

	2	1	2	4	5	6
2	2,2	1,2	2,4	2,5	2,6	
1	1,2	1,1	1,4	1,5	1,6	
4	4,2	4,1	4,4	4,5	4,6	
5	5,2	5,1	5,4	5,5	5,6	
6	6,2	6,1	6,4	6,5	6,6	

$P(\text{differs}) = \frac{18}{20} = \frac{9}{10}$ ①

$P(\text{sum of 9}) = \frac{2}{10} = \frac{1}{5}$ ①

$P(\text{differs}) = \frac{4}{10} = \frac{2}{5}$ ②

(a) $a = -\frac{1}{t^2}$

$V = t^{-1} + c$ ①

$V = \frac{1}{t} + 5$ ①

as $t \rightarrow \infty$ $V \rightarrow 5 \text{ m/s}$ which is the limiting velocity.

41) $x^2 + (x+k)^2 = 16$
 $2x^2 + 2kx + k^2 - 16 = 0$
 $\Delta = 4k^2 - 4 \cdot 2 \cdot (k^2 - 16)$
 $= -4k^2 + 128$ ①

(i) touches when $\Delta = 0$ ①
 $\therefore k = \pm 4\sqrt{2}$

(ii) Intersects when $\Delta > 0$
 $-4k^2 + 128 > 0$
 $-4\sqrt{2} < k < 4\sqrt{2}$ ①

(c) $\frac{3-x}{2} = x^2 - 2$
 $0 = 2x^2 + x - 7$
 $x+B = -\frac{1}{4}$ ①

$\therefore$ x coord of M point = $-\frac{1}{4}$ ①
 $\therefore \left(-\frac{1}{4}, \frac{13}{8}\right)$ ①